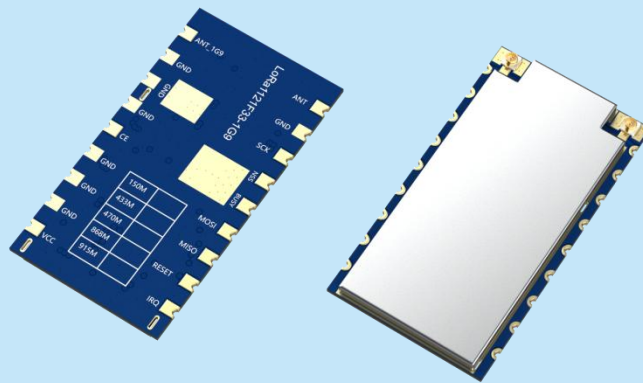


High-Power Wireless Communication Module

- Frequency: 400-480 MHz 2W, and S-band 1.9-2.0 G 1W high power
- Supports AES-128 encryption and decryption algorithms
- ANT PIN supports both stamp hole and U.FL options, allowing flexible selection for customers
- 0.5PPM TCXO crystal

Product Specification



Catalogue

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Note: Revision History

Revision	Date	Comment
V1.0	2024-11	First release
V1.1	2025-3	Update Cover,Features,Electrical Characteristics,Technical Specifications

1. Descriptions

LoRa1121F33-1G9 module is built on SEMTECH's LR1121 chip, an ultra-low power, long-range LoRa transceiver supporting Sub-GHz and satellite-connected S-band communication. The LoRa1121F33-1G9 supports LoRa, (G)FSK modulation, Sigfox protocol, and LR-FHSS. It is compatible with the LoRaWAN communication protocol and can function as a LoRaWAN node. Additionally, it offers flexible configuration options to meet the requirements of various applications and proprietary protocols.

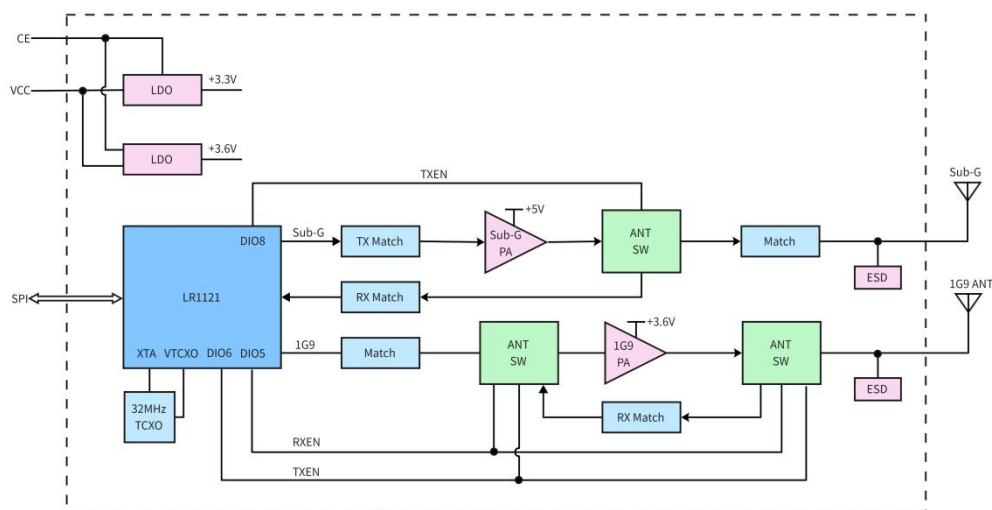
2. Features

- Sub-GHz Bands: 433/470/MHz 2W
- Sub-GHz Bands: 868/915MHz 1W
(Customizable : 150~960 MHz)
- S band: 1900MHz~2000MHz 1W
(Customizable : 2.4GHz High power)
- S band sensitivity : up to -132 dBm
@ BW=125 KHz, SF=12
- Sub-GHz reception sensitivity: up to
-144dBm @ BW=62.5 KHz, SF=12
- Electrostatic Protection (ESD)
- Supports LR-FHSS
- Supports LoRaWAN and Sigfox protocol
- Supports AES-128 encryption and decryption
- Sub-GHz transmission exceeds 10 km in
open areas
- Sleep current < 18μA
- Receive current < 9mA
- Small size, stamp hole design

3. Applications

- Drones
- Smart home/Smart agriculture
- Remote irrigation
- Industrial manufacturing

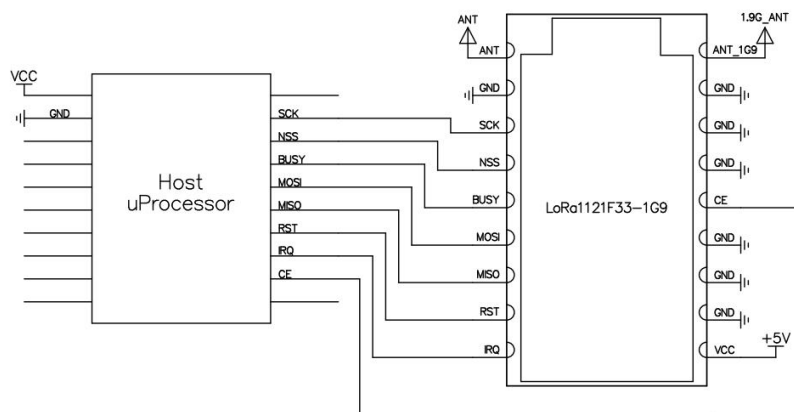
4. Block Diagram



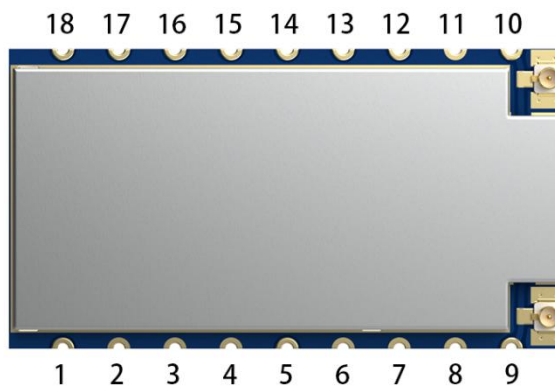
5. Electrical Characteristics

Parameters	Test condition	Min.	Typ.	Max	Unit
Voltage range		3.0	5	5.5	V
Operating Temperature		-40	25	85	°C
Maximum Input Signal			10		dBm
Current Consumption					
Transmit Current	@433MHz		< 900		mA
	@1.9GHz		< 750		mA
Receive Current			< 9		mA
Sleep Current			< 18		uA
RF Parameters					
Frequency Range	@433MHz	400		470	MHz
	@470MHz	470		510	MHz
	@868MHz	850		890	MHz
	@915MHz	900		940	MHz
Transmit Power	@Sub-GHz	20		33	dBm
	@S Band	17		30	dBm
Receive Sensitivity	BW=62.5KHz,SF=12 @Sub-GHz		-142		dBm
	BW=125KHz,SF=12 @S Band		-132		dBm
Frequency Error			0.5		ppm
Modulation Rate (@sub-GHz)	@LoRa	0.091		62.5	Kbps
	@FSK	0.6		300	Kbps
Modulation Rate (@S Frequency bands)	@LoRa	0.292		87.5	Kbps

6. Typical Schematic Circuit:



7. Pin definition



Pin NO.	Pin name	I/O	IO level	Description
1	VCC			Connect to the positive power supply
2,3,4,6,7,8,11	GND			Connect to the negative power supply
5	CE	I	0-5.5V	Internal LDO enable control pin, pull high or leave floating to operate, connect to ground for sleep mode. Internally pulled up
9	ANT_1G9			1.9GHz antenna interface, external 50Ω antenna
10	ANT			Sub-G antenna interface, external 50Ω antenna
12	SCK	I	0-3.3V	SPI clock input
13	NSS	I	0-3.3V	SPI chip select input
14	BUSY	O	0-3.3V	Used for status indication, refer to the chip datasheet for details
15	MOSI	I	0-3.3V	SPI data input
16	MISO	O	0-3.3V	SPI data output
17	RESET	I	0-3.3V	Reset trigger input, refer to the chip datasheet for details
18	IRQ	O	0-3.3V	Multipurpose digital interface, refer to the chip datasheet for details

8. Technical Specifications

1) Power Level Comparison Table (433MHz & 1.9GHz)

433MHz @SF=12, BW=125K			1.9GHz @SF=12, BW=125KHZ		
Register Value ¹	Output Power (dBm)	Transmit Current (mA)	Register Value ¹	Output Power (dBm)	Transmit Current (mA)
-5	20.1	419	-8	16.4	170

-2	22.6	427	-5	19.7	220
1	25.3	445	-2	22.9	285
4	27.2	485	1	26	380
7	29.3	551	3	28	460
10	30.8	628	5	29.1	540
13	31.7	691	7	30.0	620
16	32.2	734	9	30.2	670
19	32.6	775	11	30.3	700
22	32.9	817	13	30.3	720

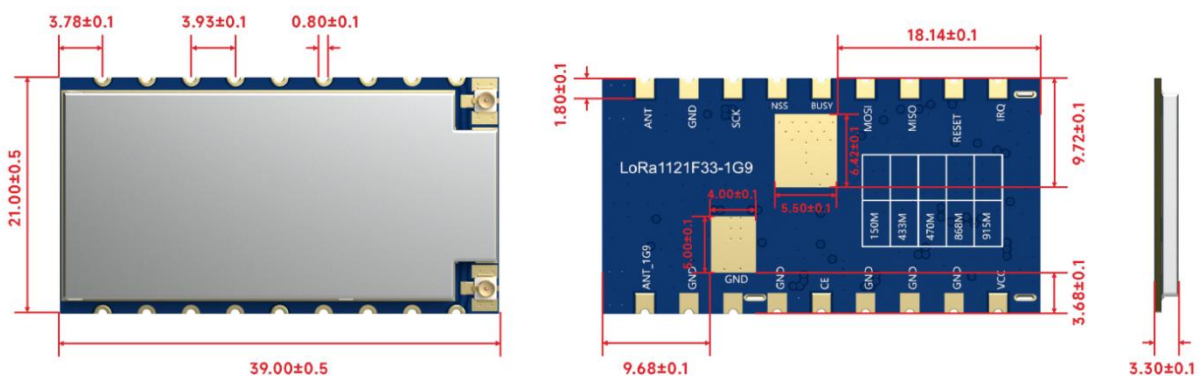
Register Value 1: TxPower in the SetTxParams(0x0211) command.

Note: For 1.9G, it is recommended not to set the TxPower above 7 for optimal efficiency. Setting the power to maximum for extended continuous transmission may potentially damage the power amplifier.

2) Voltage and Power Comparison Table

Operating Voltage (V)	433MHz		1.9GHz	
	Output Power (dBm)	Transmit Current (mA)	Output Power (dBm)	Transmit Current (mA)
3.0	28.0	506	27.0	512
3.5	29.5	604	28.0	561
4.0	30.7	678	29.6	645
4.5	31.7	736	30.3	685
5.0	32.6	810	30.3	686
5.5	33.3	844	30.3	686

9. Mechanism Dimension(Unit:mm)



Appendix 1: Function Demonstration Board



1) Function Description

The LoRa1121F33-1G9 high-power wireless module DEMO board primarily implements the following functions: inter-module communication, measuring module transmit power, measuring receive sensitivity, and measuring sleep current.

2) Key Functions

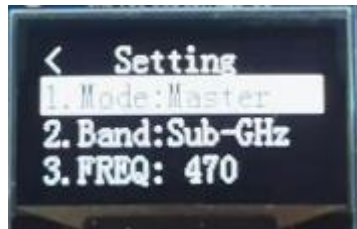
The demo board has three buttons: SET, UP, and DOWN. Their functions are as follows:

Buttons	Functions
SET Key Short Press	Confirm or enter the next level of the interface
SET Key Long Press	Return to the previous interface
UP Key Short Press	Move cursor up or increase parameter by 1
UP Key Long Press	Increase parameter
DOWN Key Short Press	Move cursor down or decrease parameter by 1
DOWN Key Long Press	Decrease parameter

Operation Method:

Ensure normal power supply, then toggle the power switch to turn on the power. The DEMO board screen will display the current function interface. Short press the SET key to enter the setting mode. To select an option, short press the SET key; to return, long press the SET key. Use the UP and DOWN keys to select the parameter you want to adjust. Finally, short press the SET key to complete the modification.

3) Setting Interface Description



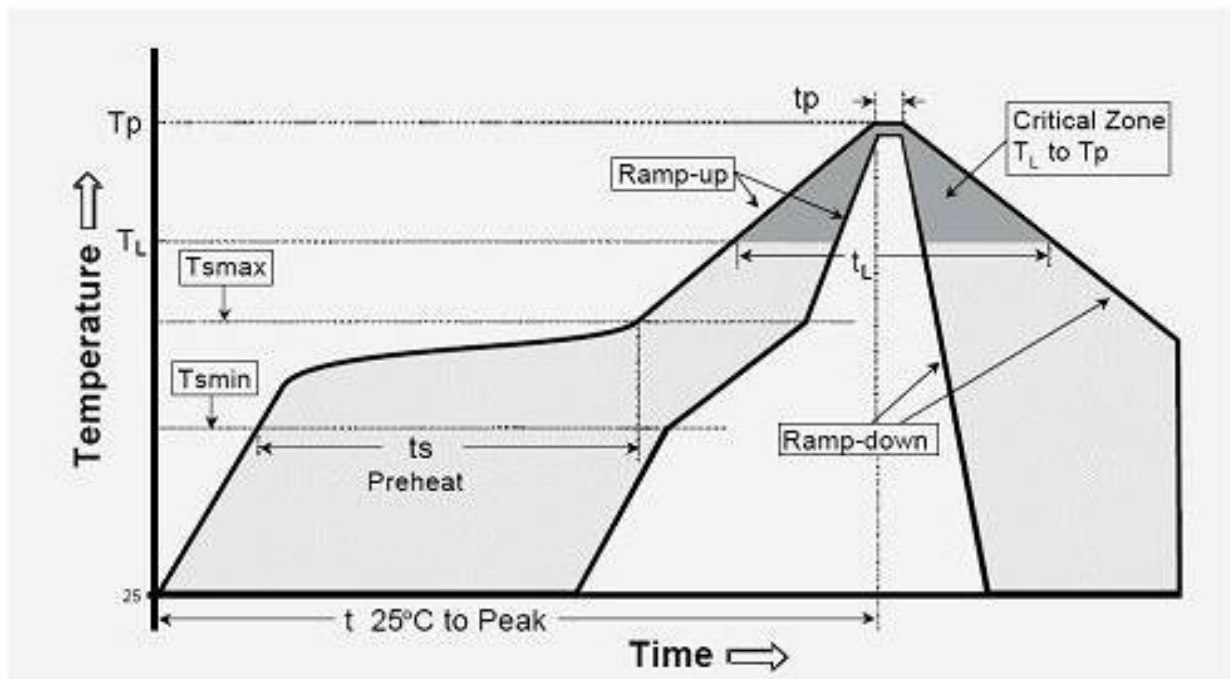
- Mode: Function
- Band: Frequency Band
- FREQ: Frequency
- SF: Spreading Factor



- BW: Bandwidth
- POWER: Power
- CR: Coding Rat

Appendix 2: SMD Reflow Chart

Below reflow profile is recommended for SMT technology:



IPC/JEDEC J-STD-020B the condition for lead-free reflow soldering	big size components (thickness $\geq 2.5\text{mm}$)
The ramp-up rate (Tl to Tp)	3°C/s (max.)
preheat temperature	
- Temperature minimum (Tsmin)	150°C
- Temperature maximum (Tsmax)	200°C
- preheat time (ts)	60~180s
Average ramp-up rate(Tsmax to Tp)	3°C/s (Max.)
- Liquidous temperature(TL)	217°C
- Time at liquidous(tL)	60~150 second
peak temperature(Tp)	245+/-5°C